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Section: C

Semester: 6th

Subject: Artificial Intelligence

AI Agent Assignment: Job Eligibility Checker

Agent Type: Simple Reflex Agent

1. Sensor

The sensor perceives the environment by collecting raw data about the applicant.

- Age of the applicant
- Educational Qualification
- Years of Experience
- Required Skills/Certifications possessed
- (Optionally) Location or Notice Period

2. Processing

Processing is the internal decision-making logic where sensor data is evaluated against predefined if-else rules to arrive at a decision.

- Age is checked against the minimum required age
- Educational qualification is compared with the required qualification
- Years of experience are checked against the minimum experience needed
- Required skills are matched against the applicant's skills
- Based on these comparisons, the agent internally decides "Eligible" or "Not Eligible"

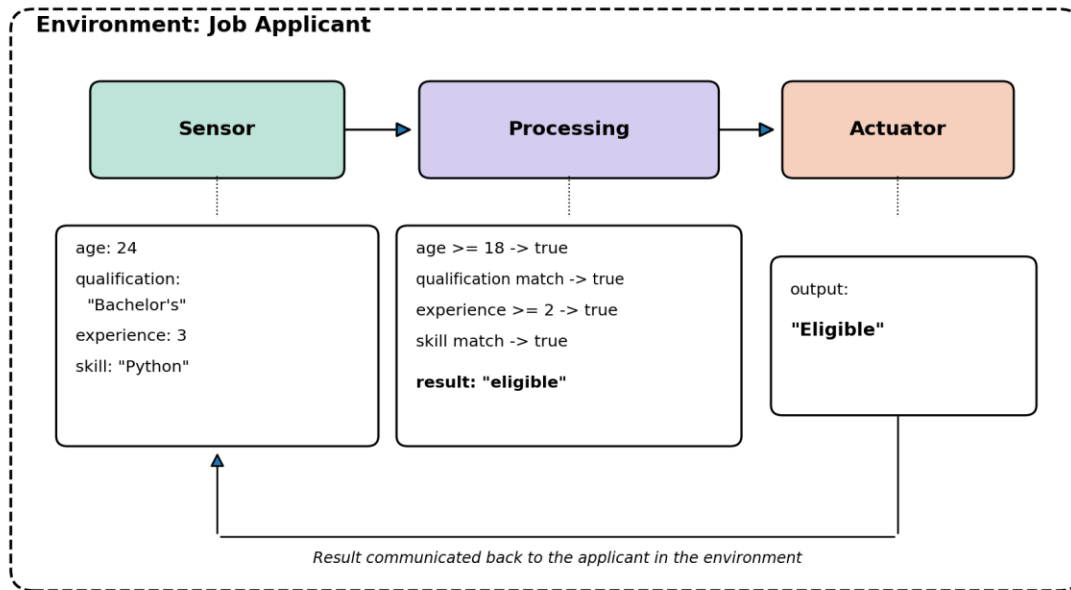
3. Actuator

The actuator performs the action based on the decision made during processing, communicating the final result to the applicant.

- If all conditions are met -> Actuator outputs: "Eligible for the Job"
- If any condition fails -> Actuator outputs: "Not Eligible for the Job"

4. Agent Diagram

(Sensor - Processing - Actuator within the Environment)



The Sensor, Processing, and Actuator operate inside the environment (the applicant). The actuator's result is communicated back into the environment, closing the perception-action loop.

5. PEAS Framework

Component	Description
Performance Measure	Accuracy of eligibility decisions, correctness of rule application, consistency of results
Environment	The applicant and the data/details submitted by them (age, qualification, experience, skills)
Actuators	Display/output unit that shows the result - "Eligible" or "Not Eligible"
Sensors	Input fields/forms through which applicant details (age, qualification, experience, skills) are collected

6. Environment Type

Property	Justification
Fully Observable	All data needed for the decision (age, qualification, experience) is directly given by the applicant. Nothing is hidden.
Deterministic	The same input always produces the same output, with no randomness involved.
Episodic	Each eligibility check is independent of previous checks.
Static	The environment does not change while the agent is processing the decision.
Discrete	Inputs and outputs come from a finite, well-defined set of possibilities.
Single-Agent	Only one agent (the checker) operates in this environment, with no other agents involved.

Conclusion

The Job Eligibility Checker operates in a Fully Observable, Deterministic, Episodic, Static, Discrete, and Single-Agent environment. Because of these characteristics, a Simple Reflex Agent using basic if-else rules is sufficient for this task, without requiring memory, learning, or handling of uncertainty.